Yifan Zhu

Waltham, MA   zhuyifan@brandeis.edu   (202)-758-8938   linkedin.com/in/yifan-zhu-b67666261
Google Scholar   kansusu.github.io

Objective

Computer Science Ph.D. candidate at Brandeis University specializing in **LLM alignment**, **Theory of Mind and epistemic reasoning in dialogue**, and **multimodal common-ground tracking**. Research investigates how humans and AI agents build, track, and repair shared understanding in collaborative interaction — combining formal epistemic modeling, multimodal perception pipelines, and beyond-preference post-training methods. Demonstrated record of leading projects from theoretical framework to deployable system and peer-reviewed publication across NAACL, AAAI, LREC, and SIGDIAL. Seeking a **research-oriented Summer 2027 internship** to apply this expertise to production-scale agent systems.

Education

Brandeis University *Expected 2027*
Ph.D. in Computer Science

- **Advisor:** Prof. James Pustejovsky, Brandeis Lab for Linguistics and Computation (LLC)
- **Research areas:** Theory of Mind & epistemic reasoning, multimodal perception, LLM alignment, computational semantics

Georgetown University *Aug 2019 – May 2021*
M.S. in Computational Linguistics

University College London *Sept 2015 – Sept 2016*
M.A. in Linguistics

Xiamen University *Sept 2011 – June 2015*
B.A. in Philosophy, Chinese Language and Literature

Research Experience

Project: Friction-Based Alignment & Theory of Mind *Oct 2025 – Present*

- Designed and implemented a suite of **friction-based LLM alignment methods** (**FAR, FTR, FPP, GRFR**) under the **Frictive Policy Optimization (FPO)** framework, training agents to issue context-aware interventions conditioned on per-turn epistemic risk rather than static preference signals.
- Introduced a **four-identifier Theory-of-Mind schema** reconstructing first- and second-order belief states at each referring expression; ablating the second-order channel cuts misunderstanding detection from 65% to 26%.
- Extended FPO from shared-context to **perceptually asymmetric dialogue**, formulating F^{-} per-perspective; perspective-bound contradiction detection on MapTask saturates repair-productive friction recall with an open-weights model where omniscient GPT-5 falls short.
- Built the full training pipeline: GPT-5 belief extraction, deterministic friction surrogates, QLoRA fine-tuning of **Qwen3.5-27B** (DPO, FAR, FPP, FTR), and a calibrated evaluation suite (F1, ECE on warranted / aligned subsets, LLM-judge debiasing).

Related publications: Mind the Gap: Theory-of-Mind-Grounded Friction for Epistemic Alignment (under review, EMNLP 2026, first author); *From Propositional to Perceptual Asymmetry: Extending FPO to Asymmetric Partial Information Dialogue* (SIGDIAL 2026, first author); *Modeling Theory of Mind in Multimodal HCI* (HCII 2024, first author).

Project: Common Ground Tracking & Epistemic Modeling *Mar 2024 – Present*

- Introduced the **Distributed Partial Information Puzzle (DPIP)** task and dataset — the first benchmark in which each participant holds a fully disjoint view of a target structure — to evaluate LLM-based and axiomatic DEL-based belief-tracking paradigms.
- Used **evidence-based dynamic epistemic logic (EB-DEL)** to model beliefs and intentions from multimodal evidence, enabling construction and refinement of common ground in collaborative interactions.
- Linked Generative Lexicon event structures with DEL via **Lexical Event Modeling (LEM)**, grounding verb meaning in executable pre-/post-state transitions to drive belief updates across multiple modalities.
- Authored **multimodal annotation guidelines** for common ground and friction labels; led adjudication and annotator training; contributed the **block-description transfer module** to the real-time TRACE common-ground demo.

Related publications: Distributed Partial Information Puzzles (LREC 2026, first author); *TRACE: Real-Time Common Ground Tracking* (NAACL 2025, co-author); *Common Ground Tracking in Multimodal Dialogue* (LREC-COLING 2024, co-author); *Lexical Event Models for Multimodal Dialogues* (HCII 2024, co-author); *Multimodal Common Ground Annotation Guidelines* (2025, lead author).

Project: Multimodal Perception & Situational Awareness *May 2023 – Present*

- Developed a **2D perception pipeline** extracting verbal and non-verbal events from audio–video recordings (posture, gesture, object movement, gaze, facial expression).
- Developed a **3D perception pipeline** extracting verbal and non-verbal events from **live-stream data**, enabling real-time analysis of multi-party interactions.
- Designed and implemented an end-to-end **multimodal situational-awareness pipeline** for collaborative problem solving; selected and applied classifiers for CPS codes via **late-fusion** integration of multimodal features.

Related publications: Multimodal Situational Awareness in Collaborative Learning (HCII 2025, first author); *Speech Is Not Enough: Interpreting Nonverbal Behavior in Multiparty Dialogue* (AAAI 2025, co-author).

Project: Computational Semantics & Cross-Lingual UMR *2022 – 2023*

- Conducted sentence- and document-level **Universal Meaning Representation (UMR)** annotation for Chinese, contributing to cross-linguistic semantic analysis.
- Identified Chinese-specific annotation challenges (verb compounds, aspect, anaphora) and proposed compound-annotation solutions adopted in the community guidelines.

Related publications: UMR Annotation of Chinese Verb Compounds (UMR Workshop, 2023, co-author).

Technical Skills

- **Languages:** Python, C/C++, Bash, LaTeX
- **ML / LLM stack:** PyTorch, HuggingFace Transformers, TRL, PEFT / LoRA / QLoRA, DeepSpeed, vLLM, W&B
- **Alignment & post-training:** RLHF, DPO, GRPO, PPO, reward modeling, KL trust-region control, offline RL
- **Evaluation & analysis:** ECE / calibration, bootstrap CIs, LLM-as-judge with bias debiasing, perspectivist annotation
- **Multimodal:** 2D / 3D perception pipelines, speech diarization, gesture / gaze / posture extraction, late-fusion classifiers
- **Infrastructure:** multi-GPU training (A100 / H100), SLURM, Docker